Problem Statement and Goals
LingoLyrics

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Table 1: Revision History

Date	Developer(s)	Change
September 22, 2026	Omar Bakr	Initial Revision 0 draft
September 28, 2026	Raymond Ma	Filled out the template; implemented supervisor feedback; reworded documentation with Codex.
September 28, 2026	Rhea Gokhale	Changed goals to selling point level, confirmed customer documents, overall polish and review

1 Problem Statement

This section describes the learning and content-authoring problem addressed by LingoLyrics, its high-level inputs and outputs, intended stakeholders, and operating environment.

1.1 Problem

Self-directed language learners at an early-intermediate to intermediate level often use songs to encounter authentic vocabulary, idioms, and cultural context. However, conventional music players and lyric websites do not help learners turn those moments into focused, in-context study. When a learner encounters an unfamiliar word or phrase, they must pause, search elsewhere for meaning, and manually transfer useful vocabulary into a study system. This interrupts the learning experience and separates vocabulary from the lyrical context needed to understand its intended meaning.

The challenge is greater for languages with non-Latin scripts, different word boundaries, or different sentence structures. Learners may require language-appropriate segmentation, transliteration, and contextual explanations in addition to direct translation. Creators also lack a consistent, reusable way to prepare synchronized lyric lessons with timing, translations, and annotations.

LingoLyrics will deliver a mobile song-based learning experience and a companion content-authoring workflow. Together, they will allow learners to study synchronized, annotated lyrics and retain selected vocabulary through Anki.

1.2 Key Terms

- **Song map:** a structured lesson file that connects an audio track with timed lyrics, translations, transliterations, annotations, and language metadata.
- **Contextual annotation:** an explanation of a word, phrase, or lyric line that captures its intended meaning in the song.
- **Segmentation:** splitting a line of text into individual words, which is needed for languages such as Japanese that do not put spaces between words.
- **Transliteration:** writing a word in another alphabet (for example, Japanese written in Latin letters) so a learner can pronounce it before they can read the original script.
- **Anki:** a spaced-repetition flashcard application used for vocabulary review.

1.3 Inputs and Outputs

Inputs:

- Audio tracks, lyric text, timed word and line boundaries, translations, transliterations, and contextual annotations.
- Learner interactions, display preferences, and selected vocabulary.

Outputs:

- A validated, reusable song map containing synchronized lyric and annotation layers.
- A synchronized lesson view with contextual support for selected words or lines.
- An optional vocabulary item, including its selected context, sent to the learner's Anki workflow.

1.4 Stakeholders

The primary stakeholders are self-directed early-intermediate to intermediate language learners, who need an engaging way to study authentic language in context, and song-map creators, who need a practical workflow for accurately synchronizing and annotating lessons.

Secondary stakeholders include:

- Anki users, who benefit from reliable vocabulary handoff;
- the project supervisor and development team;
- copyright holders and content providers, whose rights constrain what material may be used or distributed.

1.5 Environment

The learner-facing application is designed for short, self-directed study moments on an iOS or Android smartphone, including while listening to music with headphones. The creator workflow will be delivered through a web-based editor used on a laptop or desktop computer. Both workflows must use a shared song-map representation.

The initial release will use team-created, public-domain, or appropriately licensed content. It does not promise streaming-service integration, a production music catalogue, universal language support, or distribution rights for copyrighted lyrics or recordings.

2 Goals

The table below contains the goals that LingoLyrics must achieve to be considered acceptable and complete.

Language learners often have to reconstruct a song's context across separate tools, such as lyrics websites, translation tools, and music players. This interrupts listening and makes it harder to understand and retain vocabulary in its original lyrical context.

LingoLyrics combines timed lyrics, translations, contextual annotations, and vocabulary export in one workflow. Its reusable song-map format prevents learners and teachers from having to reconstruct the same context separately.

Table 2 lists the goals that LingoLyrics must achieve to be considered acceptable and complete. Each goal has a measure that will be refined into requirements in the Software Requirements Specification (SRS).

Table 2: Minimum Viable Product Goals

Goal	**Explanation**	**Reasoning**
Synchronized Lesson Playback	Across every lyric-line transition in a fixed two-song acceptance suite, at least 95% occur within 0.5 seconds of their mapped timestamps.	Timing keeps the lesson connected to the music.
In-Context Vocabulary Support	Every annotated target in the fixed two-song acceptance suite displays its translation and contextual annotation when selected.	Meaning is clearer when vocabulary is preserved in context.
Creator-to-Learner Interoperability	A song map created by the editor validates against one shared format and is opened by the learner application without manual file modification.	Learners receive consistent lessons, while creators avoid duplicate content preparation.
Practical Lesson Creation	At least two of three test creators can produce a complete map for a pre-defined short test track within 45 minutes, without manually editing the underlying map file; each resulting map must validate and play in the learner application.	A lesson system is only useful if content can be prepared consistently.
Language-Aware Support	The application must correctly display lyrics for two languages with different text-handling needs. At least one language must use a non-Latin writing system, and its word divisions and transliterations must match the song map.	This shows that the application can support different languages and writing systems without displaying text incorrectly.
Vocabulary Retention Handoff	Sending a selected vocabulary item creates a note in the named *LingoLyrics Test* Anki deck with correctly populated *Term*, *Translation*, and *Context* fields.	This connects song learning to spaced-repetition study.

For goal evaluation, the **fixed two-song acceptance suite** contains one map for each demonstrated language and is defined before implementation testing begins. An **annotated target** is a word or line with a translation and

contextual annotation. A **complete map** contains all required audio, timing, lyric, and annotation data. A **test creator** is a participant given the same supplied audio, lyric, and annotation material; the **pre-defined short test track** contains no more than 12 lyric lines.

3 Stretch Goals

The table below shows goals that are not required for the product to be viable, but would provide additional value if achieved.

Table 3: Stretch Goals

Goal	**Explanation**	**Reasoning**
Third-Language Support	Add another language and a pluggable language-processing approach.	Demonstrates that the format can scale.
Song-Map Import and Export	Add migration and interchange tooling for song maps.	Helps maps remain reusable over time.
Collaborative Review	Support sharing or review of song maps.	Encourages higher-quality learning content.
Learner Progress and Accessibility	Add progress tracking, in-app review, or accessibility enhancements.	Extends learning value beyond playback.
Timing and Annotation Assistance	Evaluate automatic assistance while preserving creator verification.	May reduce authoring effort without sacrificing accuracy.

4 Custom Documents (CDs)

- Fall: Design Thinking Report or User/Stakeholder/Client Interview Report: interviews with target learners and song-map creators to validate the problem and the learner and creator workflows before design begins.
- Winter: User Manual or Usability Report: usability testing of the mobile learner app and the web editor with representative users, measured against the goals in Table 2.

The Winter CD was changed from a User Manual to a Usability Report following instructor feedback. The fall custom document will validate learner and creator workflows. The winter custom document will document and evaluate the editor and mobile lesson flows.

Generative AI Use

The Generative AI Use Disclosure covering this document is in the appendix of the LingoLyrics Development Plan.

Appendix — Reflection

1. **What went well while writing this deliverable?**
   The team quickly aligned on the project direction and maintained open, consistent communication while preparing the document. This made it easier to discuss decisions, divide work, and keep the problem statement and goals consistent.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**
   A few meetings had to be rescheduled, which made coordination harder; we resolved this by setting recurring Friday meeting times. The harder problem was the level of detail in the goals. Our first drafts were either too vague to measure or so specific that they read like test cases (for example, naming the exact Anki deck and field names). We resolved this by checking each goal against the checklist question "is this a selling point or a requirement?", keeping the measurable part, and moving implementation detail to the SRS. We also found that parts of our drafts assumed a web app while others assumed a phone app; we agreed on a mobile learner app with a web-based editor and updated both documents to match.

3. **How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project?**
   Professor feedback identified "language learners" as too broad a target group, so we narrowed the intended users to self-directed, early to intermediate learners who use songs to study a target language. We then limited the core product to two languages, a two-song acceptance suite, and team-created or licensed content, and explicitly excluded streaming-service integration and a production music catalogue. Features we liked but could not guarantee, such as a third language, collaborative review, and automatic timing assistance, became stretch goals. Keeping one non-Latin language in the core goals keeps the project technically challenging enough for a senior design project.